Epic Games Store Giveaways: Causal Effect on Steam Player Engagement

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Abstract

Does a free game giveaway on the Epic Games Store increase or decrease engagement for the same title on Steam? Using a staggered difference-in-differences event study across 446 treated games from 2018–2025, we estimate a two-phase dynamic. In the giveaway month, Steam concurrent players rise by approximately 10–14% across TWFE, LP-DiD, and stacked DiD; the effect then reverses, with twelve-month point estimates spanning roughly 0 to −27% depending on estimator. The preferred stacked-DiD specification attenuates to near zero by the end of the window. A pre-trend test rejects parallel trends in low-traffic titles, suggesting lifecycle-decline selection inflates the medium-run negatives. We read the short-run spike as evidence of an advertising mechanism, and the medium-run estimates as an upper bound on any causal substitution effect.

Keywords: platform competition, cross-platform spillovers, video games, Epic Games Store, Steam

1. Introduction

When most gamers hear about a new video game, their first instinct is to open Steam and search for it. This reflects the social ecosystem Steam has cultivated: friend lists, community hubs, reviews, achievements, and wishlist notifications so comprehensive that it functions not only as a store but as the primary discovery and identity layer for PC gaming. Even a game distributed *entirely for free* on a rival platform is liable to drive curious players to Steam first.

This intuition recently received vivid anecdotal support. In January 2026, the CEO of New Blood Interactive shared that when *Blood West* was given away for free on the Epic Games Store over the 2025 holiday period, Steam sales for the title surged by approximately 200% on the same day (FRVR Blog, 2026). "I used to think EGS was a Marketing Black Hole," he wrote, "but turns out having your game be free on Epic is great advertising for Steam sales!" The comment went viral, sparking a Reddit thread that generated broad discussion of whether the Epic giveaway program systematically benefits Steam rather than draining it (Reddit, 2026).

These observations raise a causal question: does an Epic Games Store free giveaway increase or decrease engagement for the same game on Steam? Standard substitution logic predicts a decrease: a zero-price substitute on a rival platform should divert players away from Steam. Yet the

anecdotal evidence points in the opposite direction in the short run. The giveaway may function as an advertising shock: it generates media coverage, social-media chatter, and word-of-mouth that raise awareness of the title and drive traffic to Steam, the platform where players already hold a library, social graph, and community history.

The PC gaming platform market is a compelling context. Steam, launched by Valve in 2003, remains overwhelmingly dominant with more than 120 million monthly active users and a catalogue exceeding 50,000 titles as of 2025; its competitive moat rests not only on library size but on the thick social infrastructure that creates switching costs no price cut can replicate. The Epic Games Store, launched in December 2018 on the back of *Fortnite*'s revenue, has pursued an aggressive free-game strategy as its primary differentiation lever, giving away more than 400 games at no cost to users since launch.

Against this backdrop, this paper estimates the causal effect of EGS giveaways on Steam player engagement using a staggered difference-in-differences event study across 446 games given away between December 2018 and 2025. The identifying variation is the timing of each game's first EGS giveaway: games awaiting future giveaways serve as not-yet-treated controls for currently treated games, exploiting the rolling staggered rollout of the Epic program rather than relying on an external never-treated comparison. Engagement is measured via monthly average concurrent players scraped from SteamCharts.

The central empirical finding is a two-phase dynamic. In the giveaway month and the month after, Steam player counts rise by approximately 10–14% in log terms; the contemporaneous spike is statistically significant ($p < 0.001$) across all three estimators (TWFE, LP-DiD, and stacked DiD) and passes a randomization placebo. The effect then reverses within a few months, but how far is estimator-dependent: TWFE and LP-DiD reach roughly -11% to -27% by month twelve, while the preferred stacked-DiD specification flattens to a statistically insignificant $+0.016$. The spread brackets the medium-run causal substitution effect between essentially zero and -27%, with the true magnitude likely closer to the lower end because all three estimators absorb pre-existing lifecycle decline only imperfectly. Repeat giveaways of the same game generate no detectable additional effect.

We interpret this as advertising plus selection. The $k = 0$ spike is consistent with the giveaway announcement attracting attention to the title on Steam. The medium-run negatives are partly

contaminated by selection: the full-sample pre-period joint Wald test does not reject parallel trends at the 5% level ($F_{11,444} = 1.72$, $p = 0.067$), but the point estimates drift downward toward $k = -1$, and the low-traffic subsample does reject ($F_{11,213} = 1.89$, $p = 0.042$). Any genuine causal substitution component is therefore best read as an upper bound. The central contribution is the large-scale causal estimate of the full dynamic trajectory of cross-platform spillovers from free-game promotions (446 games, 38,641 game-month observations, seven years of history), including the sustained post-giveaway reversal that prior work has not characterized.

2. Literature Review

The most directly related work is Clarkson (2025), a doctoral dissertation whose second chapter finds that games given away by Epic see Steam traffic increase by more than 24% during the giveaway window, with concurrent rises in Steam reviews and word-of-mouth activity, and larger spillovers for titles that lean on Steam-exclusive features. The present paper differs in three respects: it estimates the full twelve-month post-treatment trajectory using a staggered DiD design across 446 games and seven years of history (rather than only the contemporaneous effect on a curated sample), uses three heterogeneous-treatment-robust estimators to bracket the causal effect, and documents both the systematic medium-run reversal and the pre-period drift that points to lifecycle-decline selection, a mechanism invisible to a contemporaneous design.

More broadly, the question fits into the literature on platform competition and multi-homing in two-sided markets (Rochet and Tirole, 2003): when two platforms primarily serve the same user base, a free promotion on one may simply reallocate engagement rather than expand the aggregate market. Tudón (2022) estimates a 0.13% own-demand response per 1% rise in friends' Steam demand, providing a network-multiplier channel through which a free copy on a rival platform can lift incumbent-platform engagement; Nabil and Ariatmanto (2025) document that Steam's content-variety and ease-of-use advantages over EGS persist despite Epic's free-game program, consistent with a durable Steam incumbency advantage rooted in its community infrastructure rather than price.

3. Data

3.1. Epic Giveaway History

We obtained a Kaggle dataset (Dongre, 2025) recording every free game offered through the Epic Games Store from December 2018 through 2025, comprising 541 giveaway events corresponding to 471 unique game titles; the remaining 70 events are repeat appearances.[1] Repeats cluster in two patterns: *distant repeats*, where a game is offered again more than twelve months after its first appearance, and *proximate repeats*, where the same game is offered within a few months or even days of the first, most notably during Epic's annual holiday giveaway batches.[2] Table 1 summarizes the giveaway structure; we retain all giveaway dates for each game to support the multi-event analysis.

3.2. SteamCharts Monthly Player Data

For each Epic giveaway title matched to a Steam App ID, we scraped monthly average and peak concurrent player counts from `steamcharts.com/app/{appid}`.[3] The overall panel spans July 2012 through February 2026, but each game's time series begins only from the month it first appeared on SteamCharts (typically near its Steam release date). The median game has 88 months of data.

3.3. Merge Pipeline

Matching Epic game names to Steam App IDs required reconciling inconsistent titling such as subtitles, punctuation, and edition suffixes; Table 2 summarizes the pipeline.[4]

3.4. Final Panel and Summary Statistics

The assembled dataset contains 38,642 game-month observations across 446 games.[5] The outcome variable is $\ln(\texttt{avg_players} + 1)$, where `avg_players` is the SteamCharts monthly average concurrent player count; the $+1$ offset accommodates zero-player months and the log transformation compresses the heavy right tail driven by a small number of AAA titles.

[1] The source dataset flags repeat appearances with a "(repeat)" suffix appended to the game name. Nine titles had this suffix incorrectly applied to their first appearance; we manually corrected these so that only genuine repeats carry the suffix.

[2] The longest gap is 57 months (*Kingdom Come: Deliverance*); the shortest are gaps of a few days inside the same holiday batch.

[3] SteamCharts renders its historical tables server-side, so standard HTTP requests suffice.

[4] RapidFuzz fuzzy string matching auto-matched 413 of 471 unique names; the remaining 58 received manual review (50 resolved, 8 confirmed absent from Steam). Fifteen matched entries are DLC or in-game content mapped to the base-game App ID.

[5] One observation is dropped by the TWFE estimator as a singleton fixed effect, leaving an effective regression sample of 38,641 game-months.

Table 3 reports summary statistics: the raw player count is highly right-skewed (mean 1,584, median 65), reflecting a long-tail indie distribution with occasional blockbusters, and the log-transformed outcome is far better behaved (mean 4.43, SD 2.36). Figure 1 compares the pre- and post-giveaway distributions of the outcome and its unconditional mean by event-time, with a sharp uptick visible at $k = 0$; Figure 2 characterises the panel's structure (pre-treatment log-player distribution, the broadly even spread of treatment cohorts across 2019–2023, and the fraction of each cohort flagged as repeat).[6]

4. Empirical Method

4.1. Identification Strategy

The identifying variation is the timing of each game's first Epic giveaway across 80 distinct calendar months spanning December 2018 through late 2025.[7] Because the Epic program has cycled continuously through a broad catalogue of genres and release-years, games awaiting future giveaways serve as natural controls for currently treated games.[8]

The key identifying assumption is parallel trends: in the absence of treatment, the log player trajectory of games in different giveaway cohorts would have followed the same path after conditioning on game and calendar-month fixed effects.

4.2. Pooled TWFE Specification

The primary specification is a two-way fixed-effects (TWFE) event-study regression on the full unbalanced game-month panel:

$$\ln(y_{it} + 1) = \alpha_i + \gamma_t + \sum_{\substack{k \in [-12, 12] \\ k \neq -1}} \theta_k \cdot \mathbf{1}[\tilde{k}_{it} = k] + \varepsilon_{it} \tag{1}$$

where α_i are game fixed effects, γ_t are calendar-month fixed effects, $\tilde{k}_{it} = \text{clip}(k_{it}, -12, 12)$ is event-time clipped to the $[-12, 12]$ window, and the omitted category is $k = -1$. $\hat{\theta}_k$ estimates the average change in log Steam players at event-time k relative to the pre-giveaway baseline, pooling first

[6] 17 games have no pre-period observations and are omitted from Panel (a) of Figure 2; the repeat-indicator series in Panel (c) is based on any repeat, proximate or distant, so it exceeds the 30-game proximate-repeat count in Table 1.

[7] For the stacked DiD and Callaway–Sant'Anna estimators these treatment months are aggregated into 29 quarterly cohorts.

[8] An external never-treated control group was not constructed because doing so would require scraping SteamCharts data for a large subset of the ∼50,000 non-EGS Steam titles without a principled sampling frame; any criterion for choosing which to include risks biasing the comparison toward stable-trajectory titles.

and repeat events per game. Standard errors are clustered at the game level. Staggered treatment across 80 distinct treatment months identifies θ_k separately from γ_t, so the omitted reference $k = -1$ suffices despite the absence of a never-treated group. Rather than restrict to the 336 full-window games, we retain all 446 by clipping event-time: $k < -12$ enters the $k = -12$ bin and $k > 12$ enters the $k = 12$ bin, so the boundary-bin estimates at ± 12 should be read as averages over all periods beyond the window. Re-estimating Equation (1) on the 336-game full-window subsample yields qualitatively identical results.

4.3. Robustness Estimators

Because treatment timing is staggered across six years, the standard TWFE estimator is potentially biased when treatment effects are heterogeneous across cohorts (Callaway and Sant'Anna, 2021). We complement it with two heterogeneous-treatment-robust event-study estimators reported alongside TWFE in Section 5, plus a Callaway–Sant'Anna cohort-ATT analysis whose construction sacrifices precision for the strongest negative-weighting protection and is therefore deferred to Appendix A, with the per-cohort decomposition in Appendix B.

LP-DiD (Dubé et al., 2023). For each event time k, LP-DiD runs a separate local-projection regression of the change in log players from $k = -1$ to k on treatment, restricting controls to games not yet treated as of calendar time $t + k$. This avoids the negative weights TWFE assigns to already-treated comparators under heterogeneous effects.[9]

Stacked DiD (Cengiz et al., 2019). For each treatment cohort g (quarter of first giveaway) we form a balanced panel of cohort-g games with event-time $k \in [-12, 12]$ alongside not-yet-treated games observed over the same calendar window, and estimate

$$\ln(y_{ist} + 1) = \alpha_{is} + \mu_{sc_t} + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[\tilde{k}_{ist} = k] \cdot \text{treated}_{is} + \varepsilon_{ist}, \tag{2}$$

where s indexes the stack, α_{is} is a unit-within-stack fixed effect, and μ_{sc_t} is a stack-by-calendar-month fixed effect that absorbs aggregate shocks within each cohort.[10] Stacked DiD with stack-

[9] The requirement that controls remain untreated through the outcome period means that at large k the eligible control pool shrinks to games whose giveaway is still many months away. All cohorts are pooled within each k-specific regression, with cohort fixed effects absorbing calendar-time differences across cohorts; standard errors are clustered by game.

[10] Interacting the event-time dummies with the treated indicator ensures β_k is identified only from the differential trajectory of treated units relative to not-yet-treated controls inside each (s, c_t) cell. Games appearing in multiple stacks are weighted by $1/n_{\text{stacks}}$; standard errors are clustered by game.

by-calendar-month FE is our preferred causal estimator because it most fully neutralizes both control-composition bias and aggregate calendar-time shocks; LP-DiD and TWFE are reported alongside it as bracketing estimates likely inflated by any pre-existing lifecycle decline.

5. Results

5.1. TWFE Event Study

Figure 3 presents the pooled TWFE event-study coefficients $\hat{\theta}_k$ from Equation (1), with 95% confidence intervals clustered at the game level. Table 4 reports selected coefficients; the full 24-coefficient table is in Appendix G. In the giveaway month ($k = 0$), $\hat{\theta}_0 = 0.101$ (SE = 0.022, $p < 0.001$), implying approximately a 10.6% increase in Steam concurrent players. The effect remains marginally positive at $k = 1$ ($\hat{\theta}_1 = 0.045$, $p = 0.047$) but turns negative and highly significant from $k = 2$ onward, reaching -0.145 (13.5% below baseline) by $k = 5$ and -0.267 (23.4% below baseline) by $k = 12$. The pre-period coefficients ($k \in [-12, -2]$) are not individually statistically distinguishable from zero but visually appear to drift downward toward $k = -1$.

5.2. Estimator Comparison

Figure 3 overlays the TWFE, LP-DiD, and stacked-DiD event-study paths and Table 5 reports key coefficients. Qualitatively all three estimators show the same two-phase pattern: a positive spike at $k = 0$ that persists into $k = 1$, a reversal at $k = 2$, and statistically significant negatives by $k = 5$. The pre-period coefficients drift downward across the window but are not individually statistically distinguishable from zero given the standard errors.

The estimators diverge sharply on the medium-run magnitude. At $k = 12$, TWFE reaches -0.267, LP-DiD -0.105, and stacked DiD a statistically insignificant $+0.016$. The contemporaneous estimates are tighter (TWFE and stacked both $+0.101$, LP-DiD $+0.131$); LP-DiD's slightly larger $k = 0$ estimate reflects that its not-yet-treated controls are EGS-selected games approaching their own giveaway and so are themselves in the steepest phase of pre-treatment lifecycle decline, mechanically inflating the DiD contrast. The systematic pattern across the medium-run window is that the more aggressively an estimator absorbs calendar-time shocks within cohort stacks (stacked DiD > LP-DiD > TWFE in that dimension), the smaller the medium-run negative becomes. The three-estimator spread therefore brackets the medium-run causal substitution effect between roughly 0 and -27%, with the preferred stacked-DiD point estimate at the less-negative end and

losing statistical significance by $k = 12$. We return to interpretation in Section 6.

Pre-trend test. The joint pre-trend Wald test evaluates $H_0 : \theta_k = 0$ for all $k \in [-12, -2]$ using the full clustered covariance matrix of the TWFE coefficient vector. On the full sample the joint Wald statistic is $\chi^2_{11} = 18.89$ with $p = 0.063$, or equivalently $F_{11,444} = 1.72$ with $p = 0.067$, failing to reject zero pre-trends at the conventional 5% level but rejecting at 10%. The high-traffic subsample is a clean failure to reject ($\chi^2_{11} = 9.04$, $p = 0.618$; $F_{11,214} = 0.82$), consistent with stable pre-trends in larger titles. The low-traffic subsample, by contrast, rejects parallel trends at 5%: $\chi^2_{11} = 20.82$, $p = 0.035$ ($F_{11,213} = 1.89$, $p = 0.042$). The pre-trend violation is therefore concentrated in lower-traffic titles, where lifecycle decline is more pronounced; we return to the implications in Section 6.

Separated specification. A separated specification distinguishing first-giveaway from repeat-giveaway effects yields a null repeat coefficient and leaves the first-event path numerically identical to the pooled $\hat{\theta}_k$; full specification, results, and interpretation are in Appendix C.

5.3. Popularity-Stratified Analysis

We split the sample at the pre-treatment median of average Steam players (59.2 players/month) into 215 high-traffic and 214 low-traffic games and estimate Equation (1) separately for each half.[11] Figure 4 presents the event-study estimates. High-traffic games exhibit a cleaner pattern: a larger and more precisely estimated spike at $k = 0$ and a statistically clean pre-period, while low-traffic games are noisier and reject parallel trends, as documented by the pre-trend Wald statistics in Section 5.2. The selection-on-decline concern is therefore more concentrated in this stratum. Otherwise, the qualitative two-phase dynamic is present in both strata.

5.4. Robustness Checks

Three additional checks confirm the main pattern. The clean-sample re-estimation drops the 30 games with a proximate repeat giveaway; the holiday-batch re-estimation drops the 73 games whose first giveaway falls in one of Epic's December "15 days of free games" batches; and a randomization placebo permutes treatment dates 200 times to construct an empirical null for the contemporaneous spike. All three leave the headline TWFE path essentially unchanged. Full results are in Appendices D, E, and F.

[11] The 17 games with no pre-period observations in our panel have no defined pre-treatment median and contribute only to the pooled specification.

6. Discussion

6.1. Interpreting the Contemporaneous Spike

The $k = 0$ and $k = 1$ positive effects are consistent with the Epic giveaway functioning as an advertising shock that temporarily lifts Steam engagement. A free-game announcement generates media coverage and social-media chatter, and because Steam is the primary discovery and community layer for PC games, the resulting awareness surge manifests as Steam activity even when players also claim the free Epic copy. The magnitude (roughly 10–14% in log players) is in the same direction as but smaller than the $\approx 24\%$ found by Clarkson (2025), who measures Steam traffic in the days immediately after the giveaway window on a sample weighted toward larger titles, whereas we measure monthly average concurrent players and include near-zero-engagement games.

6.2. Interpreting the Reversal

The sustained reversal from $k = 2$ through $k = 12$ is the most novel finding, and two mechanisms likely contribute. First, Epic's curation team may preferentially select titles whose Steam player base is already declining,[12] so part of the observed post-giveaway decline may continue a pre-existing downward trend rather than reflect a causal consequence. The evidence is suggestive: the pre-trend tests in Section 5.2 localize the concern to low-traffic titles, and the stacked estimator produces a much smaller medium-run negative than TWFE, consistent with part of the TWFE path being a selection artefact. Selection cannot, however, explain the contemporaneous spike at $k = 0$, which is robustly causal across estimators and passes the randomization placebo.

Second, players who claimed the free Epic copy may gradually shift playtime from Steam to Epic, or may forgo a Steam purchase they would otherwise have made, leaving Steam with a permanently lower post-treatment baseline.[13] The ordering of medium-run estimates across estimators (TWFE most negative, stacked DiD smallest) is consistent with both mechanisms operating jointly: if selection is the dominant non-causal driver, the stacked-DiD path is the cleanest view of the causal substitution effect, which is close to zero by $k = 12$, and TWFE and LP-DiD at $k = 12$ should be read as upper bounds.

[12] Plausibly because declining games are cheaper to license, or because publishers of fading titles are more willing to accept giveaway terms.

[13] The substitution channel predicts a reversal beginning at $k \geq 2$ rather than $k = 0$ because players need time to install and play the game they claimed.

6.3. Implications for Platform Strategy

For developers, the calculus is reasonably clear once the components are separated. Any pre-existing lifecycle decline would continue regardless of the giveaway and should not enter the decision. The substitution component imposes little real cost on the developer: a player who migrates from Steam to Epic is still playing the game, and Epic's 12% revenue cut is more favorable than Steam's 30%. The deal therefore reduces to two clearly positive terms, the upfront licensing fee and the advertising spike (our estimated +10–14% at $k = 0$), and is typically favorable for a developer whose title is declining or past peak.[14]

For Steam, the program appears to be a net positive rather than a threat. The $k = 0$ advertising spike accrues directly to Steam, and the medium-run negatives are likely upper bounds (with the preferred stacked-DiD estimate near zero by $k = 12$). The deeper observation is that even free distribution through a rival platform funnels players to Steam's community layer, consistent with the view that Steam's moat is rooted in library, friends, and community activity rather than price.

For Epic, the picture is more ambiguous and our data can only partially answer it. Epic bears the full licensing cost; the measurable engagement effects of the giveaway accrue to Steam. What Epic plausibly gains instead is EGS user acquisition: launcher installs, repeat visits, and eventual paid purchases, none of which is observable in Steam-side data. If the primary measurable effect of Epic's giveaway spending is to advertise games on a competitor's platform, whether the program is strategically rational hinges on EGS user-acquisition gains that this paper cannot test.

6.4. Limitations

The two most consequential limitations are intertwined. First, there is no never-treated group: all 446 games are eventually treated, so identification rests entirely on timing variation across cohorts (see Section 4 for why an external never-treated pool was not constructed). A well-constructed never-treated comparison could in principle sharpen identification. Second, Epic may preferentially select games already in lifecycle decline, violating parallel trends for the post-giveaway reversal. The pre-trend evidence reported in Section 5.2 is consistent with selection concentrated in low-traffic titles. The TWFE and LP-DiD medium-run estimates should therefore be read as upper bounds on the true causal substitution effect; the stacked-DiD estimate of +0.016 at $k = 12$ is the

[14] Clarkson (2025) documents that Steam reviews rise alongside players during giveaways, feeding the recommendation algorithm, and the Blood West case (FRVR Blog, 2026) suggests the purchase-conversion effect can far exceed the engagement effect.

closest to a causal estimate our design can produce but is statistically insignificant. Three further limitations apply to measurement and external validity. *Player counts vs. purchases.* SteamCharts reports concurrent players, not purchases or unique installs; a giveaway effect on purchases may be larger than, or differently timed from, the player-count effect we estimate. *DLC and bundle giveaways.* Fifteen events distributed DLC or in-game content rather than the base game; mapping these to the base game's App ID likely attenuates the estimated effect for those observations. *Genre heterogeneity.* The Epic catalogue mixes single-player titles (where the substitution mechanism is weaker) and multiplayer titles (where platform choice has ongoing matchmaking consequences); the pooled estimate conflates these. A genre-stratified analysis is deferred to future work, with the popularity split in Section 5.3 serving as a partial proxy.

7. Conclusion

Using 446 games over 2018–2025 and a staggered difference-in-differences design, this paper estimates the full trajectory of Steam player engagement around Epic Games Store free giveaways. The pattern is two-phase: a short-run advertising spike of approximately 10–14% in log players at $k = 0$–1 that is robust across estimators and a randomization placebo, followed by a medium-run reversal whose magnitude depends on how aggressively the estimator absorbs cohort-level shocks. The reversal is partly contaminated by selection on lifecycle decline: the pre-trend violation localises to low-traffic titles, and the preferred stacked-DiD estimate flattens to a statistically insignificant $+0.016$ by $k = 12$. The medium-run causal substitution effect is therefore plausibly small; Epic's giveaway program appears to function in part as inadvertent Steam advertising.

Future work could separate single-player from multiplayer titles to test whether substitution is stronger where platform choice is more permanent; study Steam sales volume rather than concurrent players to quantify commercial impact directly; and examine whether giveaways generate lasting EGS engagement using Epic's own platform data, which is not publicly available.

Tables & Figures

Table 1: Epic Games Store giveaway CSV breakdown (Dec 2018–2025)

Category	Count
Total giveaway events	541
First appearances (unique titles)	471
Repeat appearances	70
Of 63 titles given away more than once:	
Given away exactly twice	56
Given away three times	7
Of 63 repeat titles, by gap to first giveaway:	
Proximate ($\leq$12 months)	30
Distant (>12 months)	33

Table 2: Steam matching pipeline and final panel construction

Stage	Count
Unique Epic titles (input to matching)	471
Matched to a Steam app ID	463
Not on Steam: Epic exclusives	5
Not on Steam: placeholder entries	3
Of 463 matched entries:	
Unique Steam app IDs	459
Duplicate collisions (different Epic name, same app ID)	4
No SteamCharts data (HTTP 500)	13
Games in final panel	**446**
Treatment month observed in SteamCharts	421
Full $[-12, +12]$ window observed in SteamCharts	336

Table 3: Summary statistics: Epic giveaway panel

Variable	N	Mean	SD	Min	Median	Max
Avg. concurrent Steam players (monthly)	38,642	1,583.9	7,444.0	0.00	65.2	270,553
Peak concurrent Steam players (monthly)	38,642	3,214.1	14,861.2	0	171	527,652
$\ln(\texttt{avg_players} + 1)$ [outcome]	38,642	4.43	2.36	0.00	4.19	12.51
Repeat-giveaway window indicator (0/1)	38,642	0.02	0.15	0	0	1
Months of SteamCharts data per game	446	86.6	41.1	1	88	163

Notes: Game–month panel, 446 games, July 2012–February 2026. Raw player counts are as reported by SteamCharts.com. The $+1$ offset accommodates months with zero reported players. The repeat-giveaway window indicator equals 1 in months within 12 months after a game's second (or later) Epic giveaway. The bottom row is game-level ($N = 446$).

Table 4: Selected TWFE event-study coefficients (pooled specification)

k	Estimate	SE	p-value
-5	0.005	0.025	0.833
-4	-0.003	0.024	0.915
-3	-0.009	0.025	0.710
-2	-0.011	0.022	0.608
0	**0.101*****	0.022	<0.001
1	0.045*	0.022	0.047
2	-0.063^{**}	0.023	0.005
3	-0.097^{***}	0.024	<0.001
6	-0.160^{***}	0.027	<0.001
9	-0.178^{***}	0.031	<0.001
12	-0.267^{***}	0.053	<0.001

Notes: Dependent variable: $\ln(\texttt{avg_players} + 1)$. Game and calendar-month fixed effects; standard errors clustered by game ($N = 38{,}641$; 446 games). Omitted period: $k = -1$. $R^2 = 0.884$; within $R^2 = 0.008$. The overall R^2 is dominated by the game fixed effects, which absorb the large between-game dispersion in player counts. The within R^2 reflects the share of residual game-month variation in $\ln(\texttt{avg_players} + 1)$ that the event-time dummies explain; its small value indicates that the giveaway effect accounts for less than 1% of within-game, within-calendar-month variation in monthly player counts.

Table 5: Estimator comparison: selected event-time coefficients

Event time k	TWFE	LP-DiD	Stacked DiD
-4	-0.003	-0.046^{*}	-0.024
-2	-0.011	-0.026	-0.013
0	0.101^{***}	0.131^{***}	0.101^{***}
1	0.045^{*}	0.099^{***}	0.049^{**}
2	-0.063^{**}	0.000	-0.040^{*}
5	-0.145^{***}	-0.064^{*}	-0.097^{***}
10	-0.199^{***}	-0.121^{***}	-0.061^{**}
12	-0.267^{***}	-0.105^{**}	0.016

Notes: Dependent variable: $\ln(\texttt{avg_players} + 1)$. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Standard errors clustered by game. Reference period $k = -1$. TWFE uses game and calendar-month fixed effects on the full unbalanced panel. LP-DiD uses not-yet-treated controls at each period k (Dubé et al., 2023). Stacked DiD constructs cohort-specific balanced panels with not-yet-treated controls and a stack-by-calendar-month fixed effect; event-time dummies are interacted with a treated indicator so that β_k is identified only from treated units (Cengiz et al., 2019). Games appearing in multiple stacks are weighted by $1/n_{\text{stacks}}$.

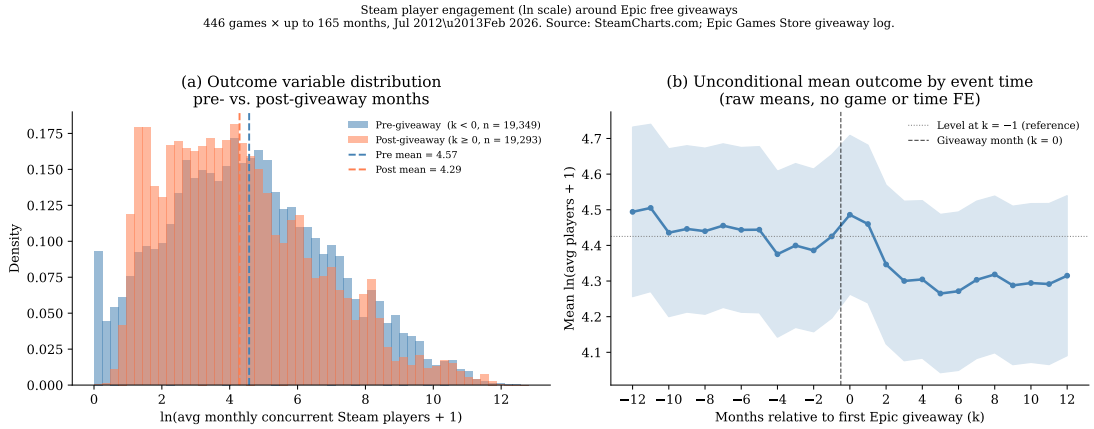

Figure 1: Steam player engagement (log scale) around Epic free giveaways. **Left:** Density histogram of $\ln(\texttt{avg_players} + 1)$ for pre-giveaway months ($k < 0$, blue) and post-giveaway months ($k \geq 0$, orange), pooled across all 446 games. **Right:** Unconditional mean (with 95% CI) of the outcome by event-time $k \in [-12, 12]$; the dashed vertical line marks the giveaway month ($k = 0$).

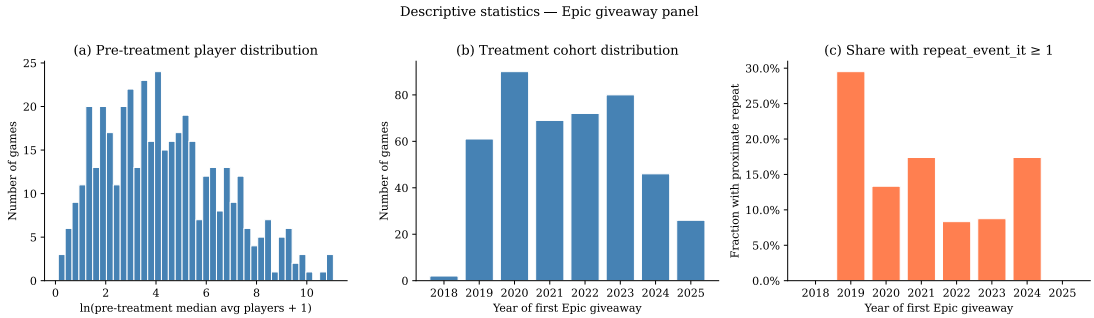

Figure 2: Panel structure and descriptive statistics. **(a)** Histogram of $\ln(\text{pre-treatment median avg players} + 1)$ across the 429 games with at least one pre-treatment observation (17 games whose first giveaway coincides with or precedes their earliest SteamCharts month have no defined pre-treatment median). **(b)** Number of games with first Epic giveaway in each calendar year (all 446 games). **(c)** Fraction of games in each cohort year that received a second or later Epic giveaway during the panel (63 games total; see Table 1).

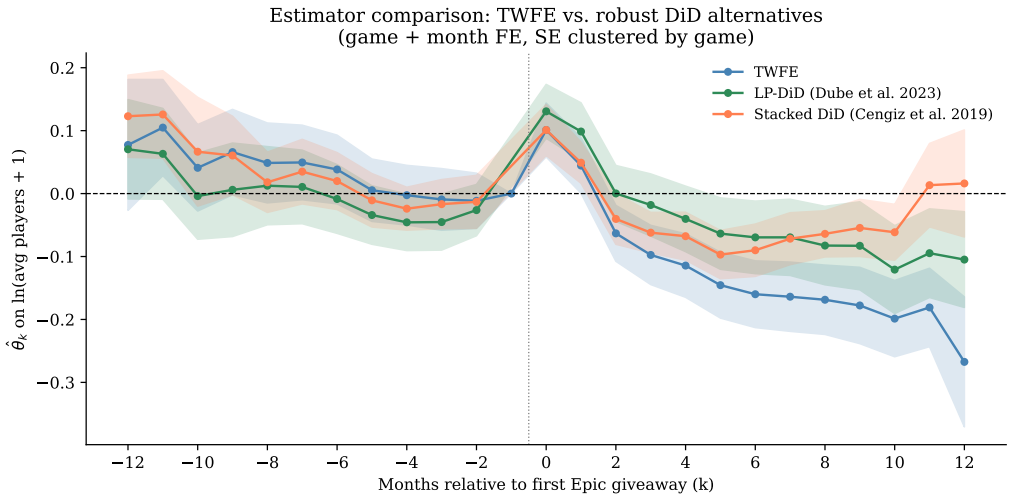

Figure 3: Event-study estimates: TWFE ($\hat{\theta}_k$, Eq. (1)), LP-DiD (Dubé et al., 2023), and stacked DiD (Cengiz et al., 2019). Outcome: $\ln(\texttt{avg_players} + 1)$. TWFE uses game and calendar-month fixed effects; standard errors clustered by game. LP-DiD uses not-yet-treated controls at each period k independently. Stacked DiD constructs cohort-specific balanced panels with not-yet-treated controls, regresses $\ln(\texttt{avg_players} + 1)$ on event-time dummies interacted with a treated indicator, absorbs unit-within-stack and stack-by-calendar-month fixed effects, and weights games appearing in multiple stacks by $1/n_{\text{stacks}}$. All estimators share the same $[-12, 12]$ event window; shaded bands are 95% confidence intervals. Omitted category: $k = -1$. $N = 38{,}641$ game-month observations; 446 games.

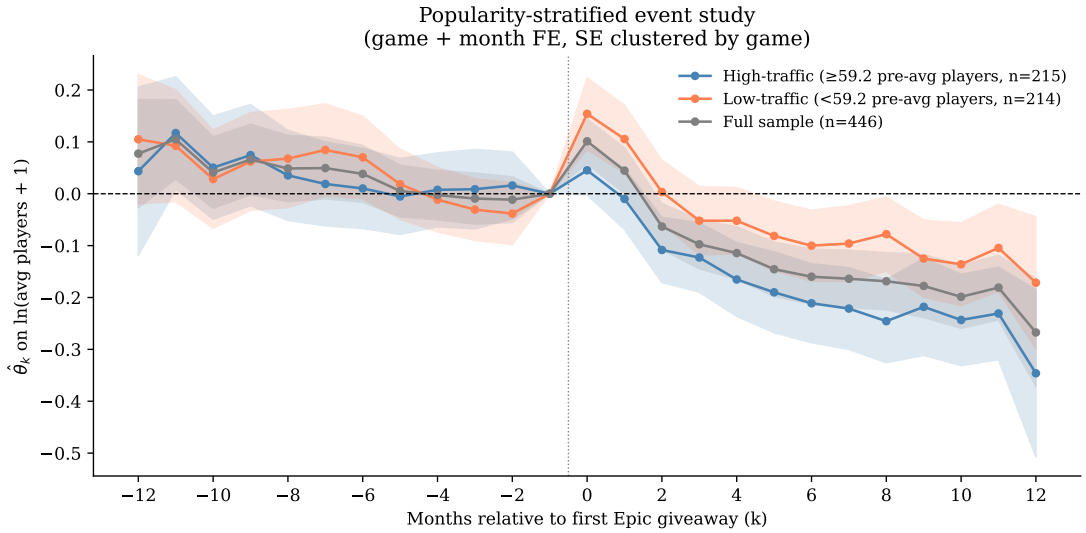

Figure 4: Event-study estimates by pre-treatment popularity. Outcome: $\ln(\texttt{avg_players} + 1)$. Split at the pre-treatment median (59.2 avg players/month): high-traffic ($N = 215$ games) and low-traffic ($N = 214$ games). 17 games with no pre-period observations are unclassified and enter only the pooled specification. Pre-trend joint Wald tests: high-traffic $\chi^2_{11} = 9.04$, $p = 0.618$; low-traffic $\chi^2_{11} = 20.82$, $p = 0.035$.

Appendix

A. Callaway–Sant'Anna Cohort ATTs

We estimate cohort-specific average treatment effects $\mathrm{ATT}(g, k)$ for each treatment cohort g (quarter of first giveaway) and event-time k, using not-yet-treated games as within-cohort controls and aggregating across cohorts weighted by cohort size (Callaway and Sant'Anna, 2021). By conditioning on narrow within-cohort comparisons, this estimator provides the strongest protection against negative-weighting bias of any specification we run, at the cost of substantially reduced precision.

Figure 5 presents the cohort-aggregated ATT estimates. The CS estimate at $k = 0$ is 0.048 (SE $= 0.040$, CI $[-0.030, 0.126]$), positive in direction but not statistically significant at conventional levels, an expected power loss given the within-cohort design. The post-period CS estimates turn negative from $k = 3$ onward, reaching -0.150 at $k = 3$ (SE $= 0.056$, $p < 0.01$), -0.143 at $k = 6$, and -0.114 at $k = 9$. Despite the reduced precision at $k = 0$, all four estimators (TWFE, LP-DiD, stacked DiD, and CS) agree on the qualitative two-phase pattern.

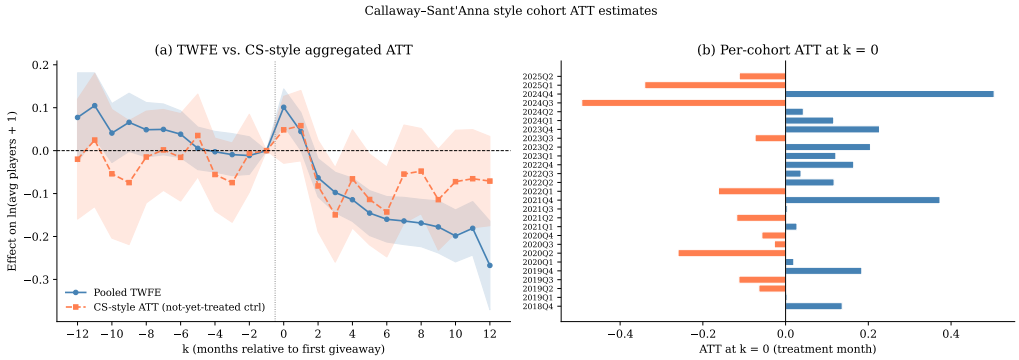

Figure 5: Callaway–Sant'Anna cohort-aggregated ATT estimates. Outcome: $\ln(\texttt{avg_players} + 1)$. Not-yet-treated games serve as controls within each treatment-quarter cohort; estimates are aggregated across cohorts weighted by cohort size. Omitted period: $k = -1$.

B. Per-Cohort Event-Study Paths

Figure 6 overlays the full event-study path $\widehat{\mathrm{ATT}}(g, k)$ for each treatment cohort, coloured from early (dark purple) to late (bright yellow) by quarter of first-giveaway exposure. Lines are semi-transparent to reveal the density of trajectories; the thick black dashed line reproduces the weighted CS aggregate from Figure 5. Individual cohort trajectories are highly dispersed and noisy: some spike sharply upward at $k = 0$, some show no visible jump, and a few drift in the opposite direction. This chaos is expected: each cohort averages only a handful of games with idiosyncratic lifecycle dynamics, so per-cohort ATTs carry wide sampling error relative to the full-sample estimate. The spike-and-decline pattern therefore emerges from averaging across cohorts rather than from any individual cohort cleanly tracing it; the per-cohort decomposition is presented here primarily to document this cohort heterogeneity.

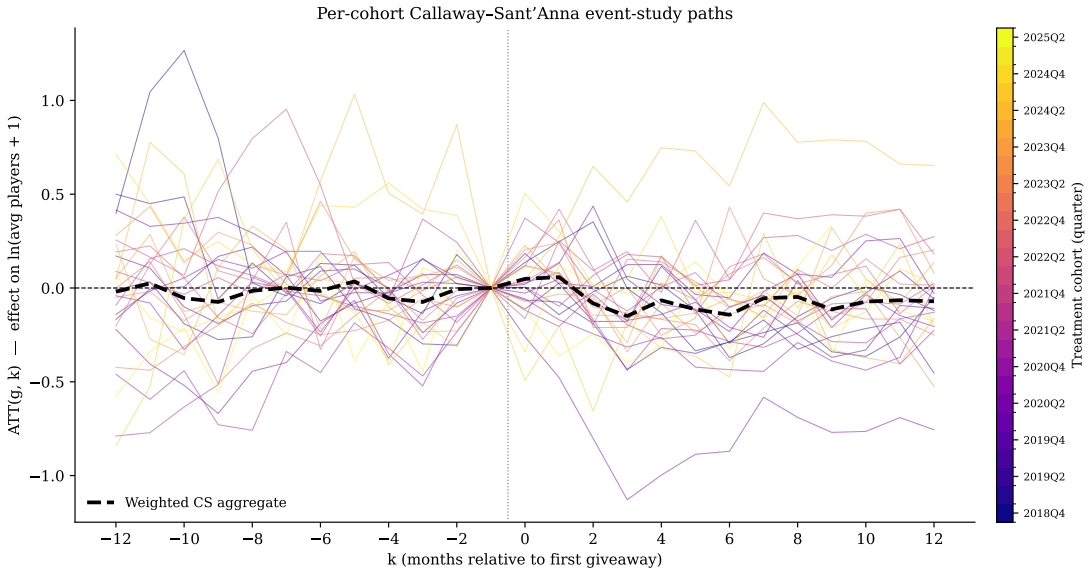

Figure 6: Per-cohort Callaway–Sant'Anna event-study paths. Each thin line traces $\widehat{\mathrm{ATT}}(g,k)$ for one treatment cohort g across event times $k \in [-12, 12]$; cohorts are coloured chronologically (early → late: purple → yellow). The thick black dashed line is the cohort-size-weighted CS aggregate. Vertical dotted line at $k = -0.5$ marks the pre/post boundary; horizontal dashed line marks zero.

C. Separated Specification: First vs. Repeat Giveaways

Specification. To separate first-giveaway from repeat-giveaway effects, we add a scalar indicator repeat_event_{it}, equal to 1 in any month t within 12 months after game i's *second or later* giveaway:

$$\ln(y_{it}+1) = \alpha_i + \gamma_t + \sum_{\substack{k\in[-12,12]\\ k\neq -1}} \beta_k \cdot \mathbf{1}[\tilde{k}_{it}=k] \;+\; \delta \cdot \text{repeat_event}_{it} + \varepsilon_{it} \tag{3}$$

β_k isolates the first-giveaway causal effect; δ captures the average incremental level shift from a repeat event holding the first-event trajectory fixed. For the 33 games with a distant repeat (>12 months after the first), $\text{repeat_event}_{it} \equiv 0$ in the event window, so δ is identified primarily by the 30 games with a proximate repeat.[15]

Results. The separated specification yields a repeat-giveaway coefficient of $\hat{\delta} = 0.0001$ (SE = 0.061, $p = 0.999$, 95% CI approximately $[-0.12, +0.12]$). The $\hat{\beta}_k$ coefficients are numerically identical to the pooled $\hat{\theta}_k$ to three decimal places, as Figure 7 makes visually clear, confirming that repeat-event contamination does not materially distort the pooled event-study path. The wide CI on δ reflects that only 63 games contribute identifying variation.

Interpretation. A zero central tendency has a clean interpretation: the marginal player who has not yet claimed or played the game after the first free promotion is a weak responder, and the awareness value of a second giveaway announcement for a title already widely known to have been free is low. The confidence interval is wide enough, however, that we cannot rule out incremental effects of ±12 log points either way. With only 63 games with any repeat and 30 with a proximate repeat, this null is an absence of evidence rather than evidence of absence.

[15] For micro-batch repeats with gaps under one month, e.g. *Escape Academy* re-offered within days during the 2023–2024 holiday batch, the cluster is collapsed to its earliest date and treated as a single event. The clean-sample specification in Appendix D drops all 30 proximate-repeat games as a robustness check and leaves main estimates unchanged.

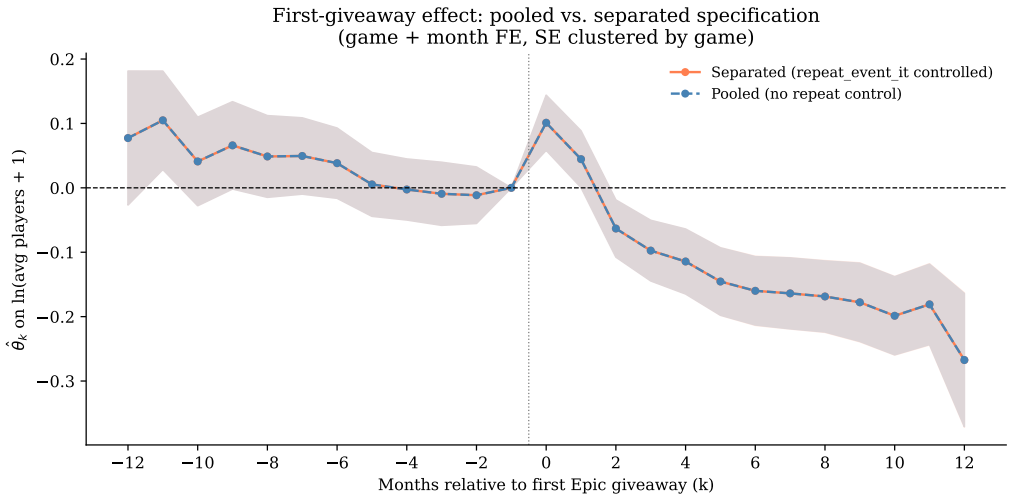

Figure 7: First-giveaway event-study path under the separated and pooled specifications. The separated $\hat{\beta}_k$ (with the repeat-event indicator partialled out) and the pooled $\hat{\theta}_k$ overlay almost exactly, confirming that repeat-event contamination does not materially distort the pooled path.

D. Clean-Sample Robustness Check

Figure 8 presents the pooled TWFE event-study on the clean sample that drops all 30 games with a proximate repeat giveaway (within 12 months), leaving 416 single-event games. The coefficients are nearly identical to the full-sample pooled TWFE (see Figure 3), confirming that repeat-giveaway contamination is not materially distorting the first-event estimates.

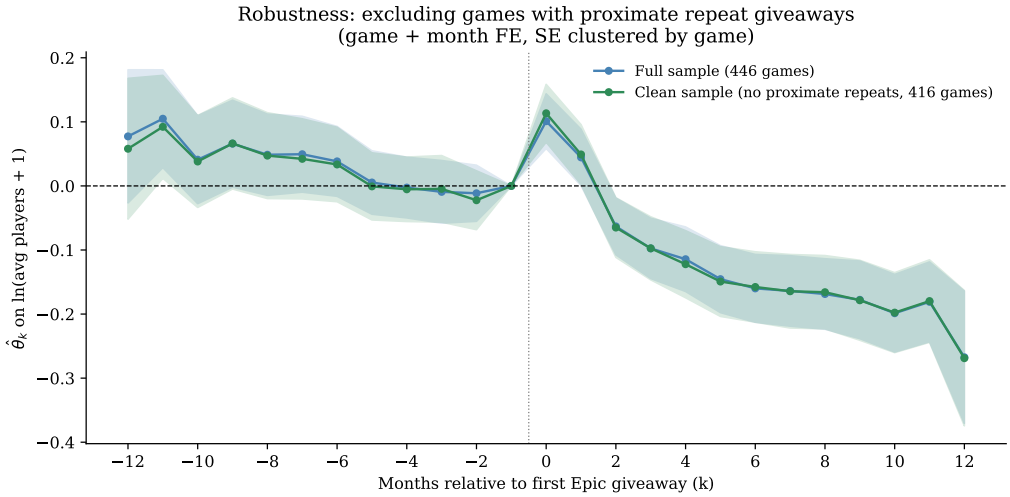

Figure 8: Clean-sample robustness check. Pooled TWFE re-estimated after dropping all 30 games with a proximate repeat giveaway (within 12 months), leaving 416 games. Coefficients are virtually identical to the TWFE path in Figure 3.

E. Holiday-Batch Robustness Check

Epic's December "15 days of free games" batches concentrate many giveaways into a single 2–3-week window of platform-wide media attention, so the short-run spike could be confounded with the surrounding holiday promotional environment. We re-estimate Equation (1) after dropping the 73 games whose first giveaway falls during one of these five December batches, leaving a 373-game sample of 32,108 game-months. The treatment-month spike is +0.109, compared with

$+0.101$ in the full sample, and the $k = 12$ estimate is -0.258, compared with -0.267. Figure 9 overlays the no-holiday event-study path on the full-sample TWFE estimates; the two paths are visually indistinguishable. The advertising spike and the medium-run decline therefore cannot be attributed to the holiday-batch media environment: they persist at similar magnitudes when those cohorts are removed.

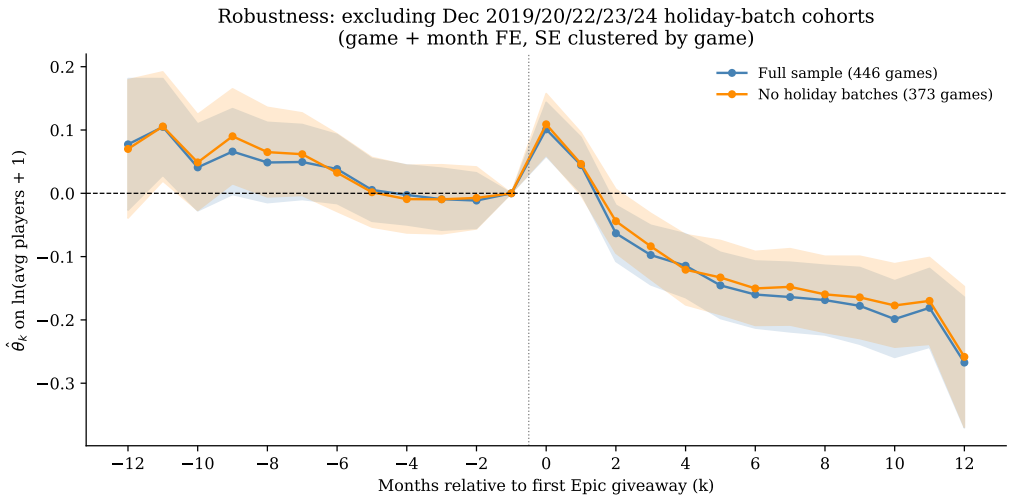

Figure 9: Holiday-batch robustness check. Pooled TWFE re-estimated after dropping the 73 games whose first giveaway falls in Epic's December holiday batches, leaving 373 games. Coefficients track the full-sample TWFE path closely.

F. Randomization Placebo

We permute treatment dates across all 446 games 200 times, preserving the empirical distribution of treatment months, and re-estimate the pooled TWFE each time. The empirical p-value compares the actual $|\hat{\theta}_0|$ to the resulting permutation null. Figure 10 shows the distribution of $|\hat{\theta}_0|$ alongside the actual estimate: the empirical p-value is $p < 0.001$, with the actual $|\hat{\theta}_0| = 0.101$ far exceeding the 95th percentile of the permutation null, 0.038. The treatment-month spike cannot be attributed to the panel's time-series structure or systematic calendar-time patterns alone.

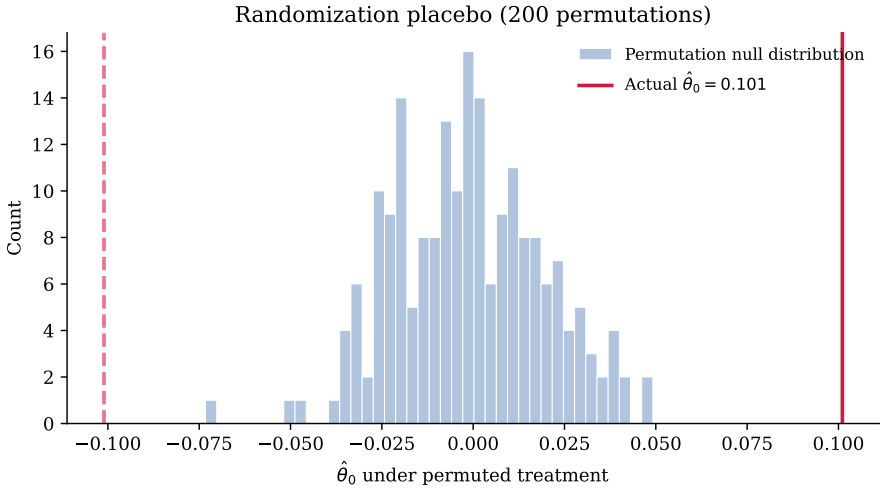

Figure 10: Randomization placebo test. Histogram of $|\hat{\theta}_0|$ from 200 permutations of treatment dates, preserving the empirical distribution of treatment months. Red vertical line: actual estimate $|\hat{\theta}_0| = 0.101$. The 95th percentile of the permutation null is 0.038; empirical $p < 0.001$.

G. Full TWFE Coefficient Table

Table 6 reports all 24 event-time coefficients from the pooled TWFE specification (Equation (1)), including confidence intervals.

Table 6: Full pooled TWFE event-study coefficients ($\hat{\theta}_k$)

k	Estimate	SE	p-value	2.5%	97.5%
−12	0.077	0.053	0.145	−0.027	0.181
−11	0.105	0.039	0.007	0.028	0.181
−10	0.041	0.035	0.242	−0.028	0.110
−9	0.066	0.035	0.057	−0.002	0.134
−8	0.049	0.032	0.132	−0.015	0.112
−7	0.049	0.030	0.102	−0.010	0.109
−6	0.038	0.028	0.168	−0.016	0.093
−5	0.005	0.025	0.833	−0.044	0.055
−4	−0.003	0.024	0.915	−0.050	0.045
−3	−0.009	0.025	0.710	−0.058	0.040
−2	−0.011	0.022	0.608	−0.055	0.032
−1	0 (ref)	—	—	—	—
0	**0.101*****	0.022	<0.001	0.058	0.144
1	0.045*	0.022	0.047	0.001	0.089
2	−0.063**	0.023	0.005	−0.108	−0.019
3	−0.097***	0.024	<0.001	−0.145	−0.050
4	−0.114***	0.026	<0.001	−0.165	−0.064
5	−0.145***	0.027	<0.001	−0.198	−0.093
6	−0.160***	0.027	<0.001	−0.213	−0.107
7	−0.164***	0.028	<0.001	−0.219	−0.109
8	−0.169***	0.028	<0.001	−0.224	−0.114
9	−0.178***	0.031	<0.001	−0.238	−0.117
10	−0.199***	0.031	<0.001	−0.259	−0.138
11	−0.181***	0.032	<0.001	−0.243	−0.119
12	−0.267***	0.053	<0.001	−0.371	−0.164

Notes: Dependent variable: $\ln(\texttt{avg_players} + 1)$. TWFE, $N = 38{,}641$; game and calendar-month fixed effects; standard errors clustered by game. $R^2 = 0.884$; within $R^2 = 0.008$.

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Note on AI usage: I used AI to assist with drafting and formatting this document in LaTeX.